

Technology may change cognition without necessarily harming it

What is the long-term impact of technological advances on cognitive abilities? We critically examine relevant findings and argue that there is no clear evidence for detrimental lasting effects of digital technology on cognitive abilities. But we also suggest how digital technology may be changing predominant ways of cognition.

Lorenzo Cecutti, Anthony Chemero and Spike W. S. Lee

From Socrates to Spielberg, every generation has worried about the next generation being ruined by the next big thing, be it writing, telephone, radio, television, computers, Internet, or cell phones. Now it is smart devices and artificial intelligence. Dovetailing with this perennial cultural narrative, a popular trend of behavioural research has linked the use of digital technology to poorer memory¹, attention², or executive functioning³. Meanwhile, digital technology is becoming ever smarter and more pervasive in our daily activities (e.g., predictive search, autonomous vehicles, social robotics). Do these observations imply that our cognitive abilities are worsening? This is a tempting inference, one that we argue against for an array of empirical and conceptual reasons.

"If men learn this, it will implant forgetfulness in their souls; they will cease to exercise memory because they rely on that which is written, calling things to remembrance no longer from within themselves, but by means of external marks." – Socrates"

"Technology... interrupts our own story, interrupts our ability to have a thought or a daydream, to imagine something wonderful, because we're too busy bridging the walk from the cafeteria back to the office on the cell phone." – Steven Spielberg"

Is digital technology worsening cognitive abilities?

The prevailing narrative that digital technology worsens cognitive abilities hinges on two assumptions: the impact is (1) about long-term abilities and (2) cognitive in nature. A critical examination of the experimental evidence suggests that the demonstrated effects have been temporary, not long-term. Their nature is also unclear; they may be about cognitive processes or about motivational factors that turn on the cognitive processes in the first place.

Consider an oft-cited series of experiments that demonstrated people's tendency to think of and rely on digital technology as a form of external memory⁴. When participants were presented with hard (vs easy) questions, technology-related words became more mentally accessible. And when participants expected information to later be accessible (vs not accessible) on the computer, they were less likely to remember what its content was, but more likely to remember where it could be found. These effects were temporary—they emerged when access to digital technology or information stored therein was available, not when access was unavailable.

Momentary effects of this sort cannot be extrapolated to infer that digital technology undermines people's long-term memory functions (or other cognitive functions). Relying on external tools when they are available is not the same as losing the ability to engage internal processes when necessary. Also, the findings indicate that access to digital technology does not indiscriminately worsen our abilities to remember, but changes the way people remember things (e.g., from what to where). As much as Socrates was concerned that people "will cease to exercise memory because they rely on that which is written," 2,000 years later, we are still capable of exercising memory—in ways attuned to our goals and situations.

Indeed, a fundamental idea in cognitive sciences is that cognitive processes operate

as required by our goals and situations, i.e., cognition is motivated⁴. The motivational value of a cognitive task influences people's likelihood and intensity of engagement in it. Digital technology may undermine the relative motivational value of a given task and thus people's engagement in it, without undermining people's ability to perform the task if they were actually engaged in it. Digital devices are particularly potent candidates for exerting such motivational influence because they are easy to access, responsive to user commands, proactive in notifying users, and capable of serving myriad goals (e.g., productivity, sociality, curiosity)⁵. By making these personally valued goals accessible and salient, digital devices may reduce relative engagement in tasks assigned by the experimenter, which, to participants, tend to be less fun, less important, or both.

For example, notifications on a smartphone or its mere presence undermined participants' performance on tasks of sustained attention², working memory, or functional fluid intelligence³. To participants, these tasks were taxing, inconsequential, and likely less motivationally valued than activities on their own smartphone, which satisfy personal goals (e.g., fitness tracking) or entertainment/escapism motivation (e.g., social media) in naturalistic settings⁵. This line of reasoning recasts the experimental findings as motivational rather than cognitive in nature. It predicts that whether digital technology hinders cognitive performance may depend on the relative motivational values of the focal cognitive task vis-à-vis activities afforded by one's smart devices.

Beyond experimental findings, there is some evidence that the more people use digital technology in daily life, the worse their performance becomes on certain cognitive tasks (e.g., associations of media multitasking with task switching⁶, episodic memory, and attention lapsing⁷). These

Box 1 | Predictions about how digital technology may be changing predominant ways of cognition as a dynamic interplay of internal processes and external tools

1. **Complexity:** Digital technology may be increasing the effectiveness (e.g., accuracy) and efficiency (e.g., speed) of performing tasks that are too complex to be handled by internal processes alone (e.g., statistical analyses, data visualization, financial accounting).
2. **Reliance and Skill:** Because of the advantages in #1, people may be (a) increasingly reliant on digital technology for computationally demanding tasks and (b) increasingly skilled at transforming information and enacting the iterative back-and-forth between internal processes and digital technology (e.g., writing code, interpreting results), skills that appear rewarded in the job market.
3. **External access → Freed capacity:** Digital technology may be expanding the scope of where and how people find information (as opposed to remembering the information itself¹), such as when they use apps that rely on machine learning to offer personalized recommendations based on their past behaviour or when they simply use Google to search for facts and GPS for driving navigation (as opposed to remembering the facts or routes), thereby freeing up their internal capacity for other cognitive tasks (e.g., intentional mind wandering, creative idea generation, learning new information).
4. **Flexibility:** People may be enjoying more flexibility in choosing what information to process internally (e.g., remembering phone numbers of significant others, remembering future intentions) and what information to rely on a smartphone for (e.g., looking up phone numbers of colleagues, using calendar reminders).
5. **Self-insights and Self-control:** People may be facing an increasing need for (a) self-insights into which cognitive activities they actually want to offload (e.g., for efficiency) and which ones they want to perform themselves (e.g., to train the internal processes they value, to retain information intrinsic to their learning goals) and (b) self-control for guarding against cognitive laziness, which is made all too easy by digital technology thanks to its provision of fingertip access to large amounts of motivationally rewarding information (e.g., reading newsfeed for hours on end).

effects are correlational, not causal, and subject to third variables and alternative explanations. Indeed, cross-sectional negative effects of smartphone use during classroom activities disappear once stable student characteristics (e.g., impulsivity, socioeconomic status) and classroom characteristics (e.g., class size, course difficulty) are controlled for in a longitudinal design⁸.

Altogether, convincing evidence remains lacking for the broad notion that technology has detrimental lasting effects on our cognitive abilities. It may, or it may not. The scientific jury is still out.

How digital technology may be changing cognition

Alongside our measured interpretation of relevant findings, we argue that technology may be changing predominant ways of cognition, without necessarily worsening it. Our argument, while speculative, is grounded in the conceptualization of cognition as it operates in complex ecological contexts—a dynamic interplay between internal processes and external tools^{9,10}.

Continuous information flow between the person and the environment is a fundamental aspect of cognition in the wild. Many of our daily activities involve offloading information and computational processes to external tools¹¹ (e.g., taking notes on paper, setting calendar reminders, doing calculations in a spreadsheet) and reloading the information and computational outcomes back to our internal processes (e.g., reading notes, receiving notifications, retrieving calculation results). Internal processes are still necessary, less for information storage and computation alone, more for first converting the information into formats that can be offloaded to external tools (e.g., converting thoughts into letters on paper, entering numbers and formulas into a spreadsheet), later reloading the information, and (along the way) managing the back-and-forth of offloading and reloading.

Through this conceptual lens, digital technology need not be viewed as a competitor for the person's internal cognitive resources; instead, it can be complementary and integral to the person's internal processes, taking care of tasks

that are difficult with internal processes alone, such as rote memorization, complex computations, and prolonged prospective memory (e.g., remembering deadlines months from now). By facilitating the person's ability to get things done via external tools, digital technology is likely to lessen the need to rely on internal processes. Accordingly, we expect digital technology to result in a number of tractable lasting changes (Box 1), which may be construed as cognitive improvements (in the dynamic view of cognition) rather than deteriorations (in the strictly internal view of cognition). In fact, positive effects of digital technology on certain kinds of cognitive performance have been found¹² (though we re-emphasize that performance may reflect motivation rather than cognitive abilities *per se*). More broadly, our predicted changes have actionable scientific implications.

Actionable implications for behavioural scientists

Extant behavioural work on digital technology has focused on how it undermines momentary, low-level cognitive processes (e.g., attention, memory). We urge fellow behavioural scientists to broaden our theorizing and research to address a series of outstanding questions raised by the dynamic perspective (Box 2). As we do so, we have to avoid drawing unwarranted extrapolations from momentary effects of digital technology to long-term ones. The latter need to be studied in their own right.

Recognizing that cognitive processes are motivated, both theoretical and empirical approaches should tease apart whether digital technology influences the relative motivational value of a given cognitive task or the actual ability to perform the task when people are motivated to engage in it. Methodologically, motivational value and engagement should be controlled for by statistical analysis, study design, or both.

Evidence for the multifaceted motivational power of digital technology and its role in problematic technology use⁵ raises the potential utility of cultivating self-control in specific technological contexts. But it is hard to train self-control with generalizable improvements in real-world settings¹³. As digital technology offers ongoing streams of novel stimuli that are often chosen algorithmically to meet customized interests and presented for passive consumption (e.g., new tweets, new videos), we are particularly worried that the younger generation, whose self-control capacity and skills are still developing, risks becoming increasingly needy for ongoing rewarding stimulation. By contrast, cognitive activities that

Box 2 | Outstanding research questions on the cognitive impact of digital technology

- If digital technology is not necessarily detrimental to cognition, then what specific internal processes does it enhance? What novel cognitive activities does it make possible?
- How are people actually using digital technology, for what cognitive work, with what frequency, in their daily life?
- What kinds of dynamic interplay between internal processes and external tools are most conducive to productivity and creativity? How are they facilitated or undermined by using digital technology? What variables predict which cognitive processes are facilitated vs undermined by digital technology (e.g., informational load, computational demand)?
- Are there demographic differences in the cognitive benefits vs drawbacks of using digital technology? For example, is digital technology more likely to improve cognition for younger people because their modes of cognitive functioning are still developing, but worsen cognition for older people because theirs are already well-formed and now require adjustment?
- What individual differences (e.g., openness to experience, goal flexibility) predict who cognitively benefits vs suffers from digital technology?
- What new forms of self-control are especially important for managing the use of digital technology?
- Developmentally, are there sensitive periods during which exposure to digital technology is particularly beneficial or harmful to a child's cognitive abilities?
- By construing the coordination of information flow between internal processes and external tools as a key component of cognition, what relevant cognitive skills should be taught and trained in schools? What aspects should be incorporated into educational testing?
- Do different platforms of digital technology (e.g., smartwatch vs smartphone vs tablet) exert systematically different cognitive effects because of the different types of activities they afford?
- If today's portable and wearable digital devices (e.g., smartwatch) constitute pervasive extensions of cognitive work, how will tomorrow's technologies (e.g., chips embeddable inside the human body) shape cognition in the wild?

lack a constant supply of custom-made rewarding stimuli and that require active internal processes (e.g., reading a novel and having to internally construct the fictive world) may become less motivationally engaging and more cognitively challenging than before. We call for research to test these possibilities and to counteract them by identifying and providing solid evidence for cognitive and behavioural strategies that are effective in messy, noisy, technology-saturated naturalistic contexts.

Such strategies, if robustly established, may then be cultivated in children and students so that they know how to manage digital technology as tools they control rather than tools that control them.

Finally, we believe that developing valid measures and educational tests of dynamic interplays between internal processes and external tools (e.g., efficient offloading and reloading of information with digital tools) may constitute a major step forward. It will help educators and

scientists identify individuals who thrive vs struggle with the use of digital technology and test ways to leverage their strengths and assist with their weaknesses. It will generate fundamental insights into whether we can use smart devices and artificial intelligence in ways that make us smarter and more intelligent. □

Lorenzo Cecutti¹ , Anthony Chemero² and Spike W. S. Lee^{1,3} 

¹Rotman School of Management, University of Toronto, Toronto, Ontario, Canada. ²Department of Philosophy and Department of Psychology, University of Cincinnati, Cincinnati, OH, USA. ³Department of Psychology, University of Toronto, Toronto, Ontario, Canada.

✉e-mail: spike.lee@utoronto.ca

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